W-R-175/1D
22 December 1986
SUPERSEDING
W-R-175/1C
15 May 1973

FEDERAL SPECIFICATION SHEET

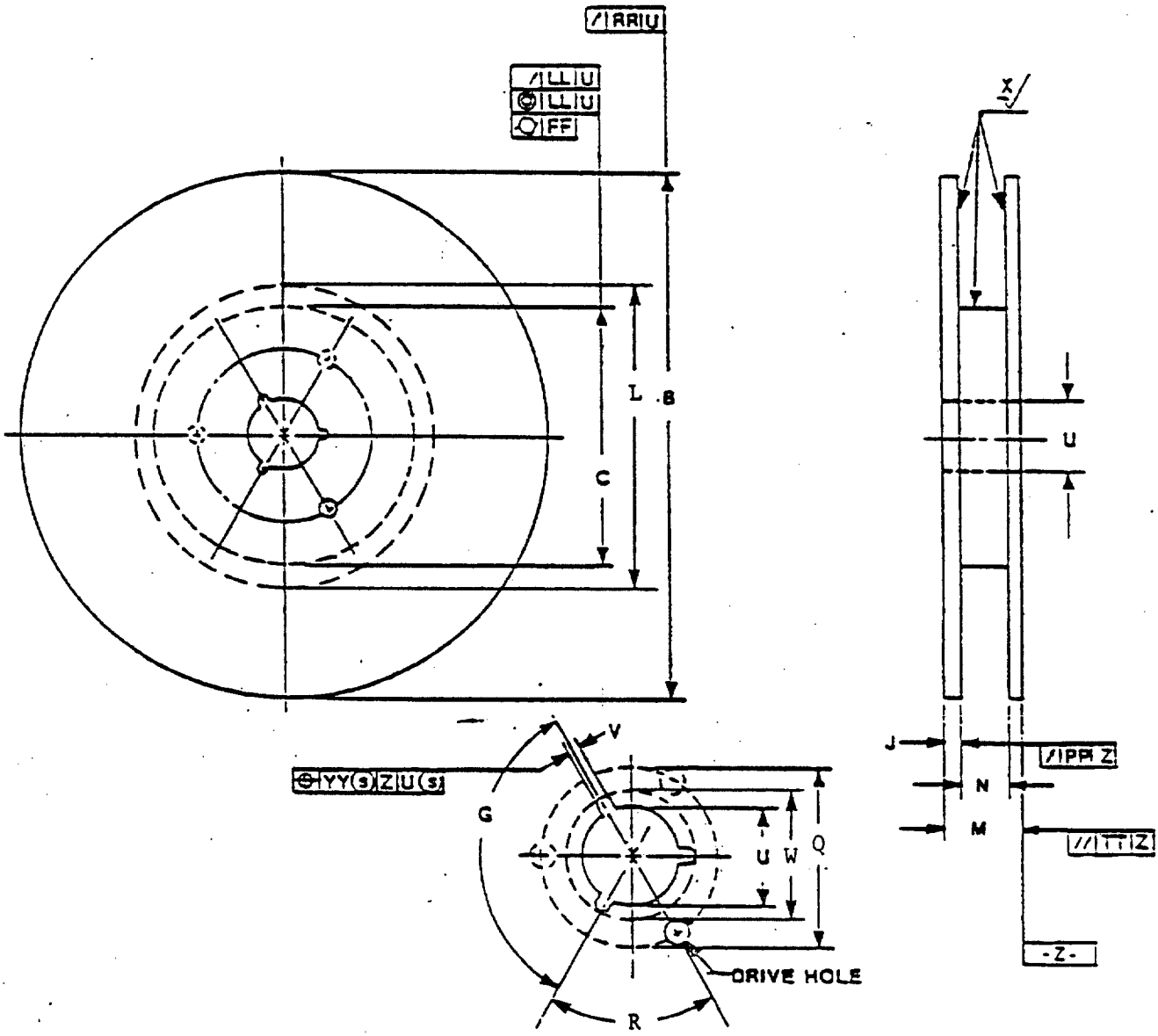
REELS, PLASTIC AND FIBERGLASS, WITH 5/16-INCH CENTER HOLE

This specification sheet is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The requirements for acquiring the reels described herein shall consist of this specification and the latest issue of W-R-175.

AMSC N/A
FSC 5835
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SH 13202770

FIGURE 1. Reel configuration.

REQUIREMENTS:

1. Materials: The reel shall be constructed of either warp-resistant plastic or fiberglass-filled plastic.

2. Dimensions: Reels shall be in accordance with the dimensions specified in table I (see figure 1).

3. Friction rings: When specified, 0.500 inch tape width reels shall be supplied with friction rings (see W-R-175/4).

4. Mounting: When specified, reel may mount from only one side.

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TABLE I. Dimensions.[1] [2]

Characteristic[3]	Reel[4] indicator	Dimensions			
		0.250-inch reel tape width		0.500-inch reel tape width	
		maximum	minimum	maximum	minimum
Outside flange dia (B)	KA	3.031 (76.99)	3.000 (76.20)		
	LA	4.031 (102.39)	4.000 (101.60)		
	MA	5.031 (127.79)	5.000 (127.00)	5.128 (130.25)	5.098 (129.49)
	NA	7.031 (178.59)	7.000 (177.80)	7.185 (182.50)	7.145 (181.48)
	PA	10.520 (267.21)	10.480 (266.19)		
Outside hub dia[5] (C)	KA	1.260 (32.00)	1.240 (31.50)		
	LA	1.760 (44.70)	1.740 (44.20)		
	MA	1.760 (44.70)	1.740 (44.20)	1.752 (44.50)[6]	1.712 (43.48)[6]
	NA	2.260 (57.40)	2.240 (56.90)	2.382 (60.50)[6]	2.342 (59.49)[6]
	PA	4.515 (114.68)	4.485 (113.92)		
Angle between successive keyslots (G) (degrees)		120.5 (2.103rad)	119.5 (2.086rad)	120.5 (2.103rad)	119.5 (2.086rad)
Flange thickness (J)		0.115 (2.921)		0.110 (2.794)	
Overall reel width (M) within lateral mounting surface		.525 (13.335)	0.490 (12.446)	.799 (20.295)	0.775 (19.685)
Distance between flanges at hub[7] (N)		.337 (8.560)	.285 (7.239)	.595 (15.113)	.571 (14.503)
Drive-hole bolt circle diameter[4] (Q)					
Flange clearance between lateral mounting surfaces, dia[13] (L)			2.250 (57.15)		2.250 (57.15)

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Dimensions			
	0.250-inch reel tape width		0.500-inch reel tape width	
	maximum	minimum	maximum	minimum
Mounting hole for plastic reels, dia (U)	0.322 (8.179)	0.316 (8.026)	0.323 (8.204)	0.317 (8.052)
Width of keyslot, plastic reels (V)	.066 (1.676)	.060 (1.524)	.083 (2.108)	.075 (1.905)
Tangent to maximum keyslot length, plastic reels, dia (W)	.640 (16.256)	.620 (15.748)	.640 (16.256)	.620 (15.748)
Mounting surface parallelism (TT)	.0025 in/in of diameter C		.0025 in/in of diameter C	
Outside hub surface cylindricity (FF)	.003 (0.076)		.003 (0.076)	
Outside hub surface concentricity (total runout) (LL)	.010 (0.254)		.010 (0.254)	
Flange rim concentricity (RR)[8]	.020 (0.508)		.020 (0.508)	
Flange wobble (PP)	.025 (0.635)		.025 (0.635)	
Keyslot position (YY)	.0028 (0.0711)		.0028 (0.0711)	
Tape path clearance[9]				
Roughness of inside flange surface[10] (X)	80 microinches		80 microinches	

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Dimensions			
	0.250-inch reel tape width		0.500-inch reel tape width	
	maximum	minimum	maximum	minimum
Roughness of out- side hub surface [10] [11]	(X) 80 microinches		80 microinches	
Flange deflection	0.17 (4.32)		0.17 (4.32)	
Open flange area (percent)[12]	50		50	

See footnotes at top of next page.

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[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] "KA" reels shall not have drive holes. "LA" through "PA" reels shall have one, two, or three drive holes. The size and shape of the drive holes shall assure that the reel fits on the gauge specified on figure 11A of W-R-175 for 1/4 inch tape width reels and the gauge shown on figure 11B of W-R-175 for 1/2 inch width tape reels. The relative position of the keyslots and drive holes shall be shown on figure 1.
[5] Lateral mounting surfaces extend out to the "C" diameter.
[6] This dimension is exclusive of any friction ring used on the reel. If a friction ring is used, it shall meet the following requirements:

(a) The material shall be neoprene of a durometer hardness of 60 ± 10 or it shall be a suitable elastomer.
(b) The outside diameter shall be concentric with the inside diameter within a tolerance of 0.020 (0.51).
(c) The inside and outside surfaces shall be cylindrical within a tolerance zone of 0.010 (0.25).
(d) The friction ring shall meet the temperature and humidty requirements of W-R-175.
(e) When mounted on the reel, the ring shall be centered between the flanges and its width shall be 0.005 to 0.008 (0.127 to 0.203) less than the applicable "N" dimension.
(f) The ring shall be flat on the hub and shall exhibit no visible buckling or puckering of the ring surface, cracking or slitting when mounted on the reel.

[7] The distance between flanges, measured at the periphery of the flanges, shall not exceed the dimension of the flanges measured at the hub (dimension "N") by more than plus 0.050 (1.270).
[8] For PA reel indicators, the flange rim concentricity shall be 0.050 (1.270).
[9] Tape path clearance shall be measured in accordance with W-R-175. The tape path center line for 0.250-inch tape shall be 0.245 (6.223), and for 0.500-inch tape the center line shall be 0.388 (9.855).
[10] For "PA" reel indicators, the inside flange and outside hub surface roughness shall be 125 microinches.
[11] Surface discontinuities resulting from the molding process, on those surfaces which are exposed to the tape and appear as a fine straight line, shall not exceed 0.0002 (0.0051) in height and 0.015 (0.381) in width.
[12] Flange holes or cutouts shall be optional as are their number, size, and shape, except that the open area shall not exceed the value shown in table I for the area between dimensions "B" and "C".
[13] "L" is the minimum diameter within which the flange, its fasteners, and/or any labeling shall not protrude outside the planes of the lateral mounting surfaces. The "L" dimension for "PA" reels shall be 4.500 (114.30) minimum.

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QUALIFICATION:

1. The qualification sample for 0.250-inch tape width "KA" through "NA" reels shall consist of ER-250-NA reels. The qualification sample for 0.500-inch tape width reels shall consist of ER-500-NA reels. The qualification sample for 0.250-inch tape width "PA" reels shall consist of ER-250-PA reels.

2. Qualification approval of ER-250-NA reels shall constitute qualification approval for all 0.250-inch tape width "KA" through "NA" reels constructed of the same material as the reels tested. Qualification approval of ER-500-NA reels shall constitute qualification approval for all 0.500-inch tape width reels which are constructed of the same material as the reels tested. Qualification approval of ER-250-PA reels shall apply to only those "PA" reels constructed of the same material as the reels tested.

3. The mounting surface parallelism and outside hub surface cylindricity measurements shall be made on completely assembled reels.

NOTE: These reels will be used with 1/4-inch wide audio recording tape and 1/2-inch wide helical scan video recording tape.

MILITARY INTERESTS:

Custodians

Army - CR
Navy - SH
Air Force - 99

Review activities

Navy - NV, OM
Air Force - 80
DLA - ES

User activity

Navy - AS

CIVIL COORDINATING ACTIVITY:
NASA - NPPO

Preparing activity:
Navy - SH
(Project 5835-0079-1)

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